1 Orthogonal Complement of a Subspace

(a)

- Zero in subset

  $\langle 0, y\rangle = 0^\top y = 0$, so $0 \in \mathcal{V}^\perp$
- Closure under addition

  Let $x_1, x_2 \in \mathcal{V}^\perp$, then for any $y \in V$,

  $$\langle x_1, y\rangle = 0, \qquad \langle x_2, y\rangle = 0$$

  Then

  $$\langle x_1 + x_2, y\rangle = \langle x_1, y\rangle + \langle x_2, y\rangle = 0$$

  Therefore $x_1 + x_2 \in \mathcal{V}^\perp$
- Closure under scalar multiplication

  Take any $x \in \mathcal{V}^\perp$ and $\alpha \in \mathbb{R}$

  $$\langle \alpha x, y\rangle = \alpha\langle x, y\rangle = \alpha x^\top y = 0$$

  Therefore $\alpha x \in \mathcal{V}^\perp$

  Hence $\mathcal{V}^\perp$ is a subspace of $\mathbb{R}^n$

(b)

$$\mathcal{V} = \text{range}(V)$$
$$\mathcal{V}^\perp = \text{null}(\mathcal{V}^\top)$$

(c)

Let v be the orthogonal projection of x onto $\mathcal{V}$, then

$$v^\perp = x - v$$

By the definition of orthogonal projection, $v \in \mathcal{V}$ and $x - v$ is orthogonal to every vector in $\mathcal{V}$. Therefore,

$$v^\perp \in \mathcal{V}^\perp$$

and

$$x = v + v^\perp$$

To prove uniqueness, suppose

$$x = v_1 + w_1 = v_2 + w_2$$

where $v_1, v_2 \in \mathcal{V}$ and $w_1, w_2 \in \mathcal{V}^\perp$. Then $v_1 - v_2 = w_2 - w_1$

The LHS belongs to $\mathcal{V}$, while the RHS belongs to $\mathcal{V}^\perp$. Hence this common vector belongs to both $\mathcal{V}$ and $\mathcal{V}^\perp$: $\mathcal{V} \cap \mathcal{V}^\perp$.

The only vector in this intersection is zero. If $z \in \mathcal{V} \cap \mathcal{V}^\perp$, then z is orthogonal to itself, so

$$z^\top z = \|z\|_2^2 = 0$$

which implies $z = 0$. Therefore, $v_1 = v_2$ and $w_1 = w_2$

So it is unique.

(d)

Using V from part (b),

$$\dim \mathcal{V} = \operatorname{rank}(\mathcal{V}), \qquad \dim \mathcal{V}^{\perp} = \dim \operatorname{null}(\mathcal{V}^{\top}).$$

By rank-nullity applied to $\mathcal{V}^{\top}$,

$$\dim \operatorname{null}(\mathcal{V}^{\top}) + \operatorname{rank}(\mathcal{V}^{\top}) = n$$

Since $\operatorname{rank}(\mathcal{V}^{\top}) = \operatorname{rank}(\mathcal{V})$, we get

$$\dim \mathcal{V}^{\perp} + \dim \mathcal{V} = n$$

(e)

Suppose $\mathcal{V} \subseteq \mathcal{U}$, and let $x \in \mathcal{U}^{\perp}$. Then x is orthogonal to every vector in $\mathcal{U}$. Since every vector in $\mathcal{V}$ is also in $\mathcal{U}$, x is orthogonal to every vector in $\mathcal{V}$. Thus

$$x \in \mathcal{V}^{\perp}$$

Therefore

$$\mathcal{U}^{\perp} \subseteq \mathcal{V}^{\perp}$$

2 Coin Collector Robot

(a)

The robot has constant unit speed in the y-direction, so

$$y(t) = t$$

In the x-direction, the mass is one, so the acceleration is the applied force. With initial position and velocity equal to zero,

$$x(t) = \int_0^t (t-\tau) f(\tau)\, d\tau$$

For the first few integer times, this gives

$$\begin{aligned}
x(1) &= \int_0^1 (1-\tau) f_1\, d\tau = \frac{1}{2} f_1 \\
x(2) &= \int_0^2 (2-\tau) f_1\, d\tau = 2 f_1 \\
x(3) &= \int_0^2 (3-\tau) f_1\, d\tau + \int_2^3 (3-\tau) f_2\, d\tau = 4 f_1 + \frac{1}{2} f_2 \\
x(4) &= \int_0^2 (4-\tau) f_1\, d\tau + \int_2^4 (4-\tau) f_2\, d\tau = 6 f_1 + 2 f_2
\end{aligned}$$

The same pattern extends to all integer times. Since f_j is applied for

$$2j - 2 \le t < 2j$$

define the matrix $A \in \mathbb{R}^{2n\times n}$ by

$$A_{tj} = \begin{cases} 0, & t \le 2j-2, \\ \frac{1}{2}(t-(2j-2))^2, & 2j-2 < t < 2j, \\ 2(t-2j+1), & t \ge 2j \end{cases}$$

Then the x-coordinate at integer time t is

$$x(t) = \sum_{j=1}^{n} A_{tj} f_j$$

Finally, the robot coordinates at integer time t are

$$\left(\sum_{j=1}^{n} A_{tj} f_j,\ t \right)$$

(b)

Since $y_i = i$, the robot can collect coin i only at time $t = i$. Using the matrix A from part (a), if $x_{\text{coin}} \in \mathbb{R}^{2n}$ is the vector of coin x-coordinates, then collecting all coins requires

$$Af = x_{\text{coin}}$$

So the desired coin vector must lie in the range of A:

$$x_{\text{coin}} \in \text{range}(A)$$

This can be checked using least squares, and checking the error is zero:

$$\hat{f} = \arg\min_{g} \|Ag - x_{\text{coin}}\|_2$$
$$r = A\hat{f} - x_{\text{coin}}$$

If the residual is nonzero, no input force vector can collect that set of coins exactly.

(c)

$$n = 6, \qquad x_{\text{coin}} = \begin{bmatrix} 0.5 & 2 & 2 & -2 & -4.5 & 0 & 13 & -4 & -6 & 12 & 22.5 & -2 \end{bmatrix}^\top$$

These are the horizontal coordinates of the 12 coins. Collecting every coin requires $Af = x_{\text{coin}}$ where $f = (f_1, \ldots, f_6)$.

From the first coin,

$$\frac{1}{2}f_1 = 0.5 \Rightarrow f_1 = 1$$

Coin 3 gives

$$4f_1 + \frac{1}{2}f_2 = 2 \Rightarrow 4 + \frac{1}{2}f_2 = 2 \Rightarrow f_2 = -4$$

Coin 5 gives so

$$8f_1 + 4f_2 + \frac{1}{2}f_3 = -4.5 \Rightarrow 8 - 16 + \frac{1}{2}f_3 = -4.5 \Rightarrow f_3 = 7$$

Coin 7 requires

$$12f_1 + 8f_2 + 4f_3 + \frac{1}{2}f_4 = 13 \Rightarrow 12 - 32 + 28 + \frac{1}{2}f_4 = 13 \Rightarrow f_4 = 10$$

Coin 8 requires

$$14f_1 + 10f_2 + 6f_3 + 2f_4 = -4 \Rightarrow 14 - 40 + 42 + 2f_4 = -4 \Rightarrow f_4 = -10$$

Collecting coin 7 requires $f_4 = 10$, while collecting coin 8 requires $f_4 = -10$. This is impossible, so the robot can't collect all coins.

(d)

For each coin, let's remove that coin's equation from the system and solve for the force vector using the remaining equations. Then I check how far the resulting robot positions are from the remaining coin positions. If the largest error is essentially zero, then that choice works.

(e)

```
Missed coin: 7
Error at missed coin: 10.000000000000014
Input force: [1.0, -4.0, 7.0, -10.0, 20.0, -35.0]
```

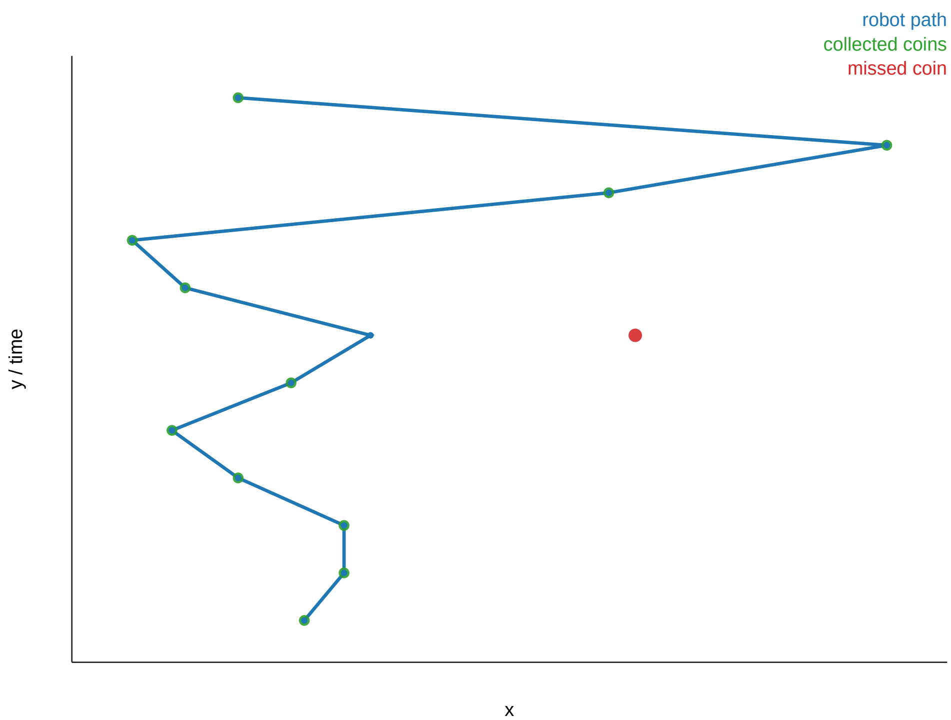

Figure 1: Coin locations and robot positions

Code

```
using LinearAlgebra

A = zeros(2n, n)
for t in 1:(2n), j in 1:n
    interval_start = 2j - 2
    interval_end = 2j

    if t <= interval_start
        A[t, j] = 0.0
    elseif t < interval_end
        A[t, j] = 0.5 * (t - interval_start)^2
    else
        A[t, j] = 2 * (t - 2j + 1)
    end
end

missed_coin = nothing
best_f = nothing
best_residual = nothing
missed_error = nothing
tol = 1e-8

for miss in 1:(2n)
    rows = [i for i in 1:(2n) if i != miss]
    f = A[rows, :] \ coin_x[rows]

```

```
    residual = norm(A[rows, :] * f - coin_x[rows], Inf)

    if residual <= tol
        missed_coin = miss
        best_f = f
        best_residual = residual
        missed_error = abs((A * f - coin_x)[miss])
        break
    end
end

println("Missed coin: ", missed_coin)
println("Error at missed coin: ", missed_error)
println("Input force: ", round.(best_f; digits=4))
```

3 Layered Medium

(a)

The equations are linear and homogeneous in the wave amplitudes, so the solution is only determined up to an arbitrary scaling. We therefore fix

$$x_1 = 1.$$

Then the scattering coefficient becomes

$$S = \frac{y_1}{x_1} = y_1.$$

The remaining unknown amplitudes are

$$x_2, \dots, x_n, \qquad y_1, \dots, y_n,$$

giving $2n - 1$ unknowns. The equations

$$x_{i+1} = t_i x_i + r_i y_{i+1}, \qquad y_i = r_i x_i + t_i y_{i+1}, \quad i = 1, \dots, n-1,$$

together with the boundary condition

$$y_n = x_n,$$

provide exactly $2n - 1$ linear equations.

Collecting the unknowns into the vector

$$z = \begin{bmatrix} x_2 & \cdots & x_n & y_1 & \cdots & y_n \end{bmatrix}^\top,$$

the equations can be written in the form

$$Az = b.$$

Solving this linear system yields all wave amplitudes, and the scattering coefficient is then

$$S = y_1.$$

(b)

For $n = 20$, $t_i = 0.96$, and $r_i = 0.02$, the computed scattering coefficient is

$$S = 0.4981789631969839.$$

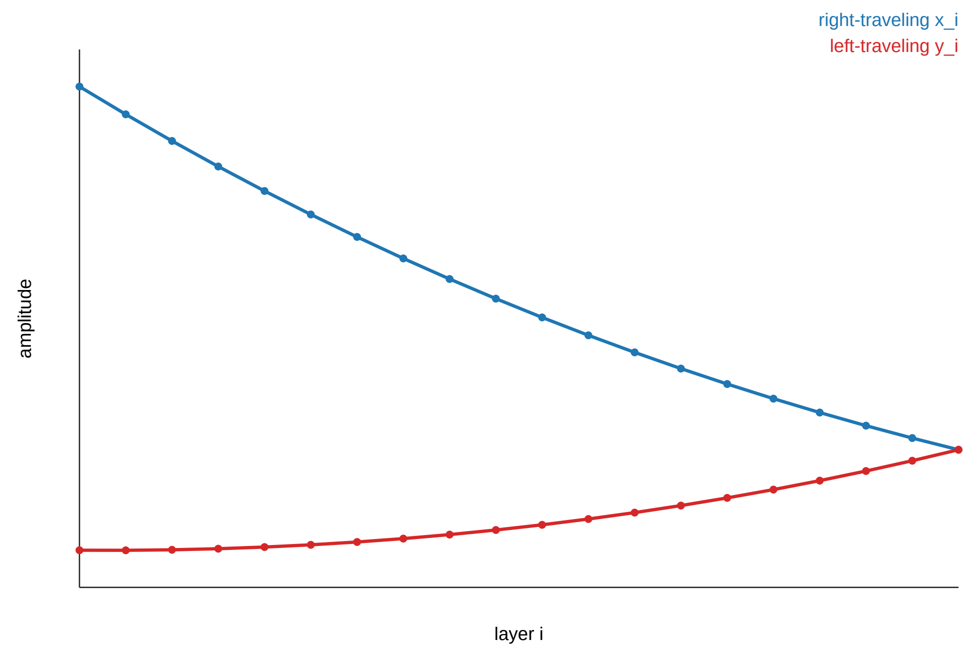

Figure 2: Right- and left-traveling wave amplitudes for the nominal medium.

(c)
For each candidate k, set

$$t_k = 0.02, \qquad r_k = 0.96$$

while keeping all other interfaces at the nominal values

$$t_i = 0.96, \qquad r_i = 0.02$$

Then solve the same linear system as in part (a), producing a coefficient S_k. The most likely fault is the k minimizing

$$|S_k - S_{\text{fault}}|$$

```
k = 9
Predicted S  = 0.7010126608436581
Error = 0.001012660843658142
```

Code

```
using LinearAlgebra

function solve_medium(t, r; x1=1.0)
    n = length(t) + 1
    nvars = 2n - 1
    A = zeros(nvars, nvars)
    b = zeros(nvars)

```

```
    idx_x(i) = i - 1
    idx_y(i) = (n - 1) + i

    row = 1
    for i in 1:(n - 1)
        A[row, idx_x(i + 1)] = 1.0
        if i == 1
            b[row] = t[i] * x1
        else
            A[row, idx_x(i)] = -t[i]
        end
        A[row, idx_y(i + 1)] = -r[i]
        row += 1

        A[row, idx_y(i)] = 1.0
        if i == 1
            b[row] = r[i] * x1
        else
            A[row, idx_x(i)] = -r[i]
        end
        A[row, idx_y(i + 1)] = -t[i]
        row += 1
    end

    A[row, idx_y(n)] = 1.0
    A[row, idx_x(n)] = -1.0

    z = A \ b
    x = [x1; z[1:(n - 1)]]
    y = z[n:end]
    return x, y, y[1] / x1
end

n = 20
t = fill(0.96, n - 1)
r = fill(0.02, n - 1)

x, y, S = solve_medium(t, r)
println("Nominal scattering coefficient S = ", S)

S_fault = 0.70
best_k = 0
best_S = 0.0
best_err = Inf

for k in 1:(n - 1)
    tk = copy(t)
    rk = copy(r)
    tk[k] = 0.02
    rk[k] = 0.96

    _, _, Sk = solve_medium(tk, rk)
    err = abs(Sk - S_fault)

```

```
    if err < best_err
        best_k = k
        best_S = Sk
        best_err = err
    end
end

println("k = ", best_k)
println("Predicted S = ", best_S)
println("Error = ", best_err)
```

4 Last Year's Midterm

1. (a) True. Since $x^T A^T A x = \|Ax\|^2$, if $A^T A = 0$, every column of A has zero norm.

(b) True. $A^T A$ is symmetric. A symmetric upper triangular matrix must be diagonal.

(c) False. For example, $A = 2I$ gives $A^T A = 4I$, which is diagonal, but A is not orthogonal.

(d) False. For example,
$$A^T A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$
is possible and has all off-diagonal entries negative.

2. $S = \text{range}(U)$. Since $U^T U = I$, UU^T is the orthogonal projector onto range(U).

3. $S \cup T$ is a subspace only if $S \cap T = S$. A union of two subspaces is a subspace only when one subspace is contained in the other; since $\dim(S) < \dim(T)$, this means $S \subseteq T$.

4. True. If $ABx = 0$, then $Bx \in \text{null}(A)$, so $Bx = 0$. Since null$(B) = \{0\}$, $x = 0$.

5. False. Let
$$A = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$
Then $\text{null}(A) \cap \text{null}(B) = \{0\}$, but $AB = 0$.

6. False. Additivity alone does not imply homogeneity over all real scalars for an arbitrary function.

7. True. Take U to be rotation by angle $2\pi/n$. Then $U^n = I$, and no smaller positive power is the identity.

8. False. The recursive construction is missing the factor $1/\sqrt{2}$ at each step. Without it, the block matrix has columns with norm $\sqrt{2}$, not 1.

9. (a) False. For example, $u = v = e_1$ gives $A = e_1 e_1^T$, which is upper triangular.

(b) True. The sum of the diagonal entries is $\text{trace}(uv^T) = v^T u = u^T v$.

(c) Q and u are parallel but may point in opposite directions. Each nonzero column of A is a scalar multiple of u, and the sign depends on the scalar from v.

10. True. $n-1$ independent vectors in $\mathbb{R}^n$ span an $(n-1)$-dimensional subspace, which is a hyperplane.

11. False. The set is the points closer to b than to the origin:
$$\|x - b\|_2 \le \|x\|_2 \iff b^T x \ge \frac{1}{2}\|b\|_2^2$$
This is a halfspace. If $b \neq 0$, the origin is not in the set, since $b^T 0 = 0 < \frac{1}{2}\|b\|_2^2$. Therefore the set is not a subspace.

12. True. If a solution exists, then there are infinitely many, since $\dim \text{null}(A) \ge n - m > 0$. Otherwise there are no solutions.

13. (a) $A \Leftrightarrow B$. The rank of $\begin{bmatrix} A & B \end{bmatrix}$ equals the rank of B exactly when the columns of A lie in range(B).

(b) $A \not\Leftrightarrow B$. A vector outside range(A) need not be orthogonal to the range, and $y = 0$ is in range$(A)^{\perp}$ but also in range(A).

(c) $A \Rightarrow B$. Infinitely many solutions imply a nonzero nullspace direction. The converse can fail if y is not in range(A).

14. Interpret $T(h)$ as the dimension of the span of the range of h. Since f and g are linear maps, this is the rank of the corresponding matrix.

(a) False. For example, take

$$F = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \qquad G = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

Then

$$FG = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \qquad GF = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

so $\text{rank}(FG) = 1$, but $\text{rank}(GF) = 0$.

(b) True. $\text{rank}(F^2) \le \text{rank}(F)$, and $\text{rank}(F + F) = \text{rank}(2F) = \text{rank}(F)$.

(c) False. $\text{rank}(FG) \le \text{rank}(G)$.

(d) True. Choose g to be the identity map. Then $f \circ g = g \circ f = f$.

(e) False. If $f = 0$, then $T(f \circ g) = 0$, so it cannot equal $T(g) > 0$.

(f) False. The same $G = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ is nonzero but nilpotent:

$$G^2 = 0$$

(g) True. If $T(f) \ge T(g)$ for every g, then f has full rank and is invertible, so $T(f \circ f) = T(f)$.

(h) True. When $m > n$, such an f has full column rank, so it has a left inverse h with $h(f(x)) = x$.

(i) False. A full-column-rank map $f : \mathbb{R}^n \to \mathbb{R}^m$ with $m > n$ is not onto $\mathbb{R}^m$, so it cannot have a right inverse onto all of $\mathbb{R}^m$.

(j) False. If $T(f) \le T(g)$ for every g, then $f = 0$. Hence $\|f(x)\| = 0$, never greater than $\|x\|$.